GRA6561 Midterm Presentation

Student ID: 1026202

BI Norwegian Business School

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Outline

1. **Question 1: Descriptive analysis**
   - Data and contract selection
   - Distributional properties and volatility
   - Trading intuition (outright vs. spread)
2. **Question 2: Predictive signals**
   - Signal design and economic intuition
   - Out-of-sample performance
   - Robustness across contracts and 2026 test
3. **Question 3: Ridge regression model**
   - Predictor construction and estimation
   - Forecast-based spread evaluation
4. **Question 4: Option pricing and hedging**
   - Black-76 pricing
   - Delta hedge and realized PnL

1.1: Selected investment universe and data treatment

We select the following six contracts:

- EEXFM1
- EEXFM2
- EEXFQ1
- EEXY1
- ENOFM1
- ENOY1

Why this selection?

- Includes required EEXFM1 and Nordic exposure
- Covers German and Nordic markets
- Includes monthly, quarterly, and yearly maturities
- EEXFM2 is relevant for the later option-pricing task

Data treatment

- Roll-adjusted daily price changes are used. On roll dates, where `SymbolKey` changes, the price change is set to zero. This avoids artificial jumps from contract replacement.

1.2: Summary statistics of roll-adjusted daily price changes

Contract	Period	Mean	Std. Dev.	Skewness	Kurtosis	JB test
EEXFM1	2017–2020	-0.02	0.99	0.29	8.88	Rejected
EEXFM2	2017–2020	0.00	0.80	0.42	6.47	Rejected
EEXFQ1	2017–2020	-0.01	0.76	0.36	6.23	Rejected
EEXY1	2017–2020	0.02	0.63	0.13	4.96	Rejected
ENOFM1	2017–2020	-0.02	0.99	0.19	7.22	Rejected
ENOY1	2017–2020	0.01	0.58	0.03	5.28	Rejected
EEXFM1	2023–2025	-0.20	3.50	-0.09	8.47	Rejected
EEXFM2	2023–2025	-0.25	3.29	-0.11	8.89	Rejected
EEXFQ1	2023–2025	-0.21	3.28	-0.29	8.92	Rejected
EEXY1	2023–2025	-0.16	3.03	-0.93	13.36	Rejected
ENOFM1	2023–2025	-0.14	2.93	-0.91	14.66	Rejected
ENOY1	2023–2025	-0.05	1.29	-1.38	14.89	Rejected

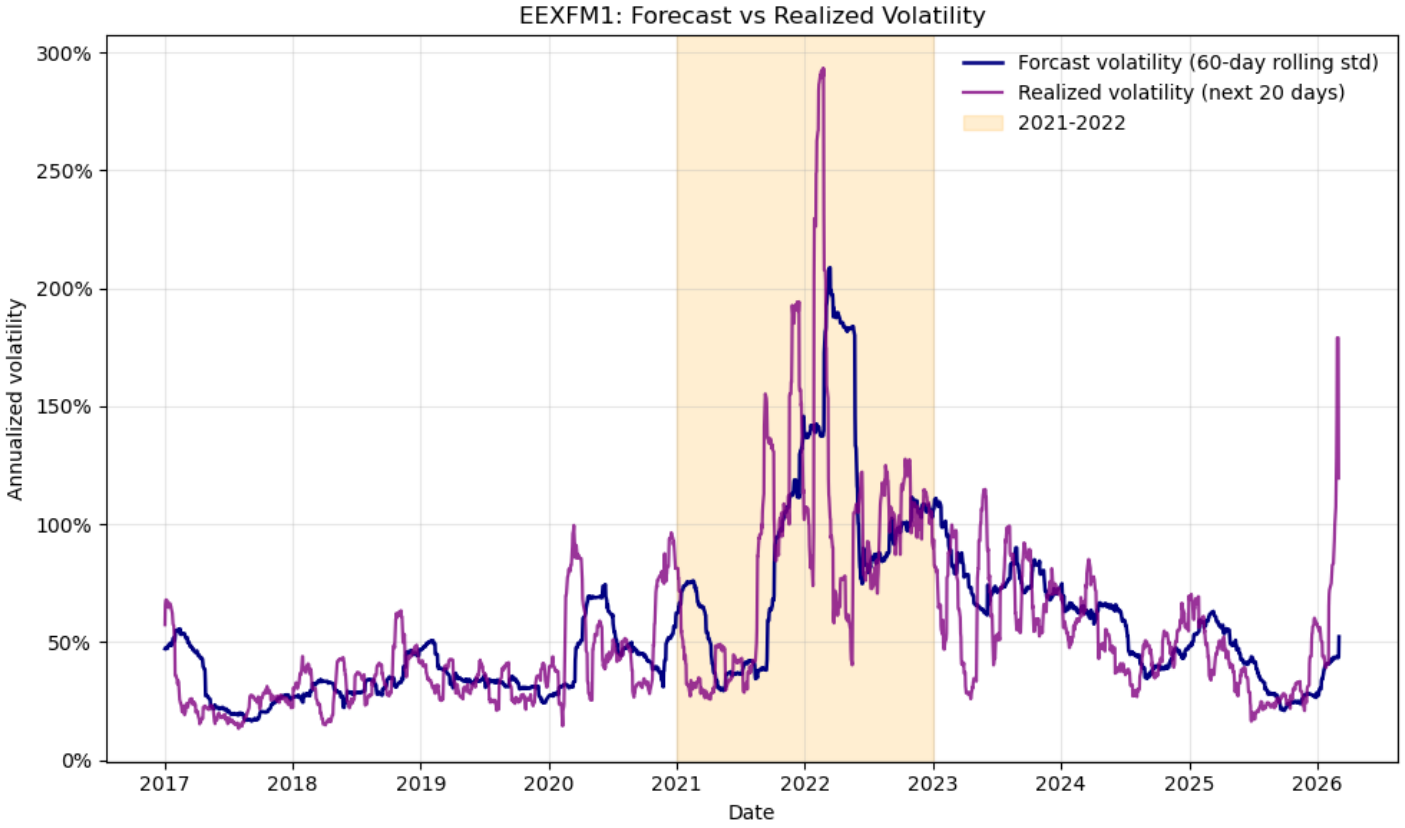
Normality tested using the Jarque–Bera statistic:

$$JB = \frac{n}{6}\left(S^2 + \frac{(K-3)^2}{4}\right)$$

where S is skewness and K is kurtosis. Under normality, $S = 0$ and $K = 3$.

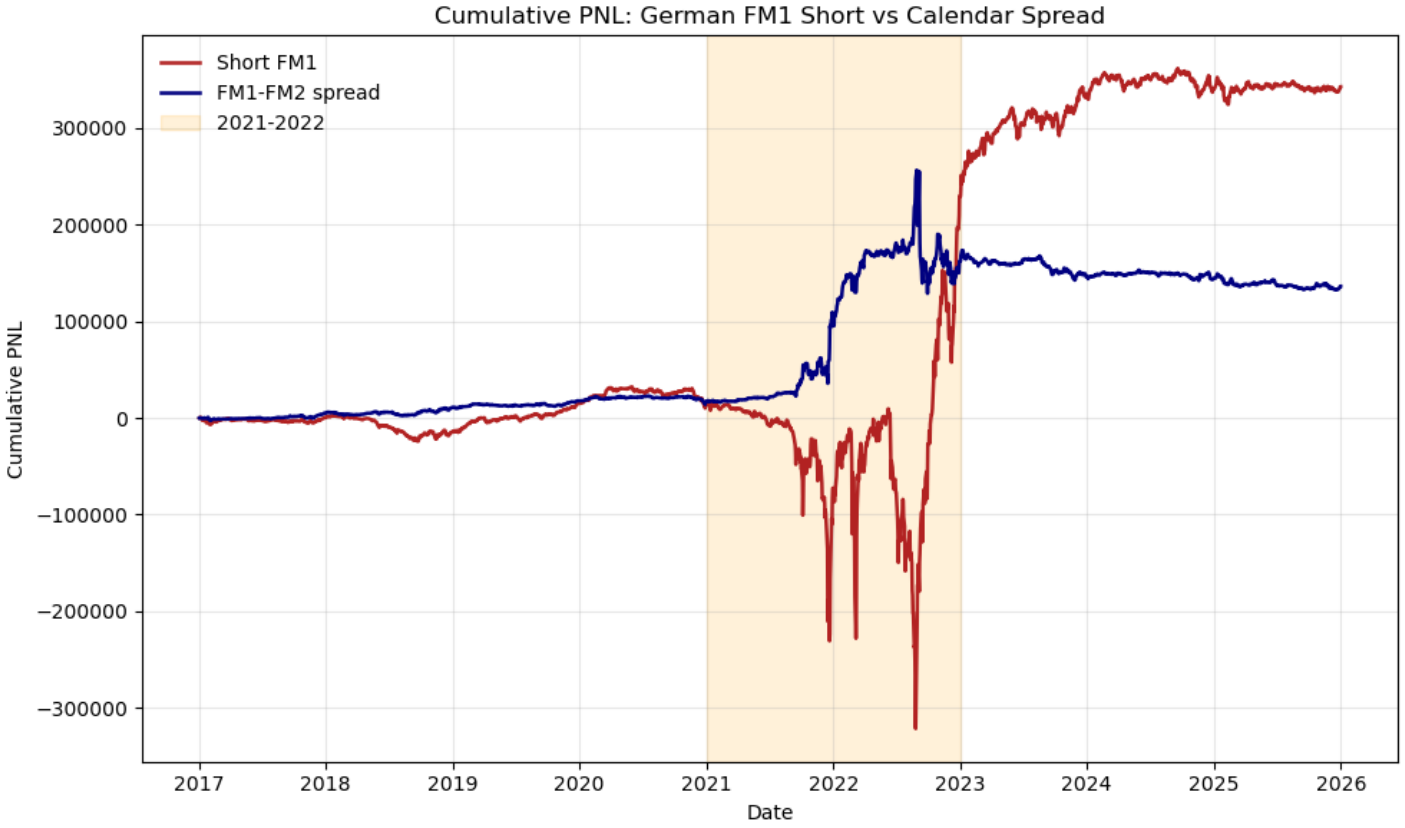
- Volatility is higher in 2023–2025
- All series remain clearly non-normal
- Several contracts show negative drift in 2023–2025

1.3: Volatility Forecasting (EEXFM1)



- **Chosen estimator:** 60-day rolling standard deviation
- Volatility is persistent and clustered, with long calm periods interrupted by sharp spikes
- 2021–2022 shows a major spike; the forecast captures it, but with some lag

1.4: Cumulative PnL: Short FM1 vs. Calendar Spread



- **Short FM1:** profitable over the full sample, but highly exposed to sharp price spikes
- **FM1–FM2 spread:** much smoother PnL path and materially lower downside risk
- The spread sacrifices some upside, but provides better protection in stress periods

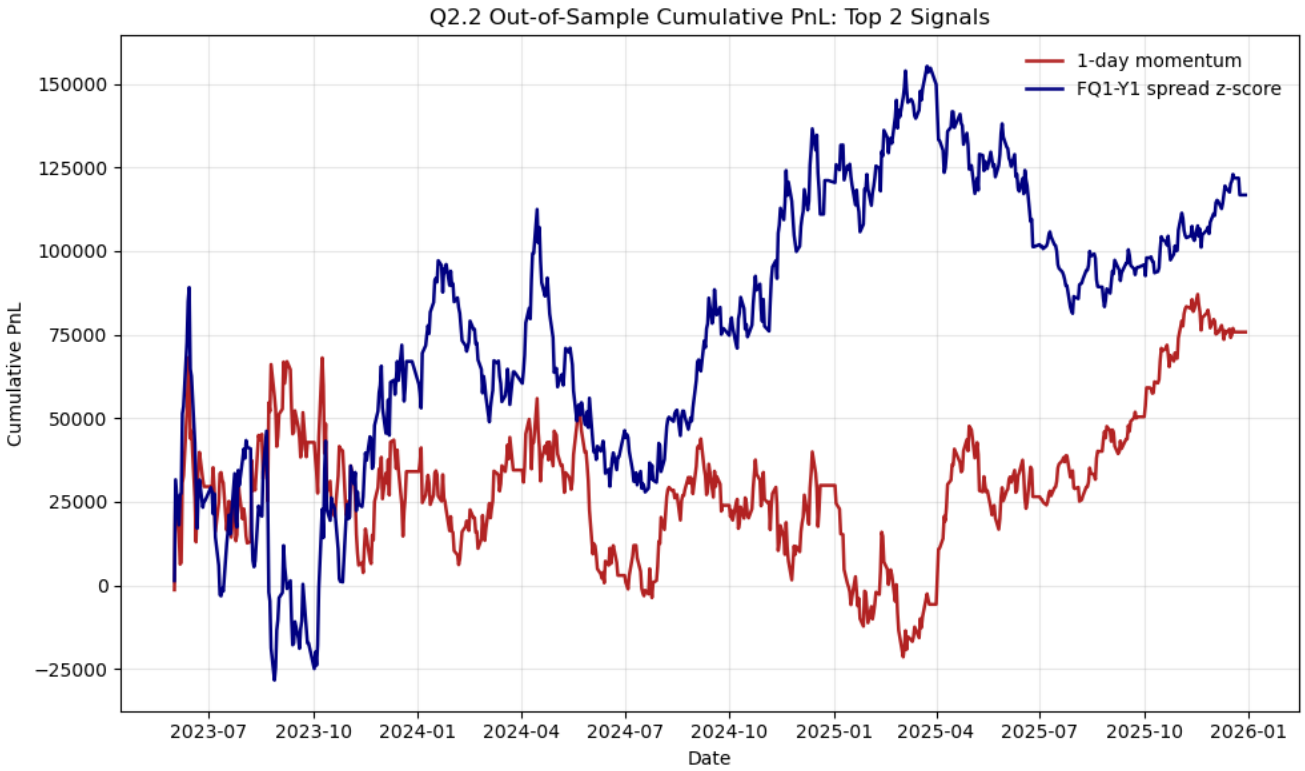
2.1: Five predictive signals (EEXFQ1)

Contract choice: EEXFQ1 (front quarter) – smoother than front month, but still liquid.

Signal	Economic intuition	Hypothesis
1-day momentum	Gradual information adjustment can create short-run persistence	Trend
5-day momentum	Unfinished weekly price moves may continue	Trend
20-day MA deviation	Distance from trend can signal pressure or imbalance	Trend / deviation
20-day z-score	Extreme deviations may reverse	Mean reversion
FQ1–Y1 spread z-score	Relative mispricing in the quarter–year spread may correct	Relative-value deviation

Signal direction is chosen in-sample using the sign of $\hat{\beta}$.

2.2: Out-of-sample test of the two best signals



Signal	Sharpe	Hit rate	Cum. PnL
1-day momentum	0.32	49.5%	75,735
FQ1–Y1 spread z-score	0.48	50.3%	116,741

Interpretation: Both signals remain profitable out of sample. The FQ1–Y1 spread z-score performs better, with higher cumulative PnL and Sharpe. Results are positive, but the Sharpe ratios are still modest.

2.3: 1-day momentum on the other five contracts

Chosen signal: 1-day momentum

Chosen because it is simple and directly comparable across all contracts. Unlike the spread signal, it does not require a benchmark contract, which makes the cross-contract test cleaner.

Contract	Sharpe	t-stat	Hit rate	Cum. PnL
ENOY1	2.28	3.93	50.3%	1,180,291
ENOFM1	2.01	3.32	52.4%	179,303
EEXY1	1.32	2.28	48.9%	1,661,668
EEXFM2	0.56	0.94	51.4%	61,509
EEXFM1	-0.09	-0.15	49.9%	-10,171

Key takeaway: The signal is profitable in 4 of 5 contracts and performs best for the year contracts. However, a substantial share of the gains in ENOY1 and EEXY1 is earned early in 2023.

2.4: 2026 holdout test and conclusion

Chosen contract: EEXY1 **Chosen signal:** 1-day momentum

Period	Sharpe	t-stat	Hit rate	Cum. PnL	Max DD
2026-01-01 to 2026-03-04	1.76	0.72	52.4%	59,743	-54,750

Conclusion: The 2026 holdout test is positive, which supports the earlier evidence that 1-day momentum works best on EEXY1. Across Questions 2.2–2.4, the signal remains profitable out of sample and generalizes better to longer-dated contracts than to the very front of the curve. However, the 2026 sample is short and the drawdown is large relative to realized profit, so the evidence should be viewed as supportive rather than conclusive.

3.1: Predictor Set, Sample Split and X Diagnostics

Predictor set

- 20 lagged predictors: 10 return-based and 10 volatility-based.
- Return signals capture short- and medium-term momentum/reversal.
- Volatility signals capture volatility level, downside risk, and term structure.

Sample split

- Train: 2017–2020 Validation: 2023–2024 Test: 2025
- 2021–2022 is omitted as a temporal gap.

X diagnostics

- No missing values after imputation; predictors scaled to $[-1, 1]$.
- X is full rank, but predictors are highly collinear.
- This motivates Ridge regularization rather than unpenalized OLS.

3.2: Ridge Estimation

Ridge solves

$$\min_{\beta_0,\beta} \sum_{i=1}^{n}(y_i - \beta_0 - x_i'\beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

where the intercept is not penalized. With $X_b = [\mathbf{1}, X]$, the closed-form solution is

$$\hat{\theta}_\lambda = (X_b'X_b + \lambda P)^{-1} X_b' y, \qquad P_{00} = 0.$$

```
def ridge_fit(X, y, lam):
    Xb = np.c_[np.ones(len(X)), X]
    P = np.eye(Xb.shape[1]); P[0,0] = 0
    return np.linalg.solve(Xb.T @ Xb + lam*P, Xb.T @ y)

grid = []
for lam in np.r_[0, np.logspace(-6, 4, 41)]:
    b  = ridge_fit(X_train, y_train, lam)
    yh = np.c_[np.ones(len(X_val)), X_val] @ b
    grid.append((lam, np.mean((y_val - yh)**2)))

lam_star  = min(grid, key=lambda z: z[1])[0]
beta_star = ridge_fit(X_train, y_train, lam_star)
yhat_test = np.c_[np.ones(len(X_test)), X_test] @ beta_star
```

- ‘Xb‘ adds the intercept column; ‘P[0,0]=0‘ excludes the intercept from the penalty.
- ‘np.linalg.solve‘ solves $(X_b'X_b + \lambda P)\theta = X_b'y$, i.e. the Ridge normal equations.

3.3: Can the model construct an optimal spread?

Ex-ante signal from Q3.2

- Ridge selected a very large penalty ($\lambda = 10000$), indicating heavy shrinkage.
- Validation R^2 was essentially zero and test R^2 was slightly negative.
- Hence, before constructing a spread, we already expected weak out-of-sample trading performance.

Q3.3 check

- We form a daily spread in 2025 by going long the contract with the highest predicted return and short the one with the lowest.
- The result confirms the ex-ante concern: cumulative return $= -21.1\%$, Sharpe $= -0.61$, max drawdown $= -31.9\%$.

Conclusion The model can generate a forecast-based spread, but the results do not support calling it an optimal spread.

4.1: Black-76 European call on the April futures contract (EEXFM2)

$$C_0 = e^{-rT}\left[F_0 N(d_1) - K N(d_2)\right]$$

$$d_1 = \frac{\ln(F_0/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

Inputs:

- F_0: April futures price on Feb 27
- $K = 70$
- T: time to expiry (Feb 27–Mar 31)
- r: risk-free rate
- σ: futures volatility

Assumptions: European exercise, lognormal futures price, constant r and σ, no arbitrage.

4.2: Price and Delta Hedge

Inputs on 27 Feb 2026

- $F_0 = 73.67$, $K = 70$, $T = 32/365 = 0.0877$
- $r = 2\%$, $\sigma = 35.25\%$, contract size $= 720$

Black-76 output

$$C_0 = 5.17 \text{ per MWh} \qquad \Delta = 0.705$$

$$\text{Call value per contract} = 3719.9$$

Hedge decision

- Selling one call gives option delta -0.705
- Delta hedge: go **long 0.705 April futures contracts** per call sold
- Equivalent futures exposure: **507.4 MWh**

Interpretation: Since $F_0 > K$, the option is in the money and has a relatively high delta, so a short call requires a long futures hedge.

4.3: Numerical Delta and Gamma

Central-difference formulas

$$\Delta \approx \frac{C(F+h) - C(F-h)}{2h}, \qquad \Gamma \approx \frac{C(F+h) - 2C(F) + C(F-h)}{h^2}.$$

Code

```
def delta_num(F, h):
    return (C(F+h) - C(F-h)) / (2*h)

def gamma_num(F, h):
    return (C(F+h) - 2*C(F) + C(F-h)) / h**2
```

Diagnostic

- We compare the numerical Greeks with the closed-form Black-76 Greeks over a grid of step sizes h
- The best step size is $h = 0.01$
- At $h = 0.01$, the errors are 6.26×10^{-8} for Δ and 3.11×10^{-9} for Γ

Interpretation

- If h is too large, discretization error increases
- If h is too small, round-off error increases, especially for Γ

4.4: Realized PnL on 3 Mar 2026

Setup

- The underlying is kept fixed as the April-2026 futures contract.
- The call is repriced on 2026-03-03 using the close price, updated time to expiry, and updated realized volatility.

Case	Option PnL	Hedge / Approx.	Total PnL
No hedge	-18,821	0	-18,821
Delta hedge	-18,821	13,666	-5,156
Delta-gamma approx.	-18,821	25,342	6,521

Comment

- The April futures price jumped from 73.67 to 100.60, so the naked short call lost heavily.
- Delta hedging reduced the loss materially.
- The delta-gamma line is a **second-order Taylor approximation**, not a fully traded gamma hedge with a separate option instrument.